

GATE QUESTION IN 2008 Q08

Question

Q.8

The minimal sum-of-products expression for the logic function f represented by the given Karnaugh map is

K-map:

PQ \ RS	00	01	11	10
00	0	1	0	0
01	0	1	1	1
11	1	1	1	0
10	0	0	1	0

- (A) $QS + PR\bar{S} + PQ\bar{R} + PRS + PQR$
 (B) $\bar{Q}S + PRS + PQ\bar{R} + PRS + PQR$
 (C) $PR\bar{S} + PQ\bar{R} + PRS + PQ\bar{R}$
 (D) $\bar{P}RS + PQ\bar{R} + PRS + PQ\bar{R}$

Answer: (A)

Explanation:

Analyzing the given Karnaugh map:

- We identify adjacent 1s and group them into the largest possible clusters (octets, quads, pairs).
- The largest groups formed are:
 - A group covering RS = 01 with PQ = 00, 01, 11 $\rightarrow QS$.
 - A group covering PQ = 01, RS = 11, 10 $\rightarrow PR\bar{S}$.
 - A group covering PQ = 01 and 11 with RS = 01 $\rightarrow PQ\bar{R}$.

- A group covering $PQ = 11, RS = 01, 11 \rightarrow PRS$.
- A group covering $PQ = 11$ with $RS = 01 \rightarrow PQR$.
- Combining these terms gives:

$$QS + PR\bar{S} + PQ\bar{R} + PRS + PQR$$

which matches option (A).